# CAPSTONE PROJECT MANAGEMENT REPORT

## Farmers Market Square E-Commerce Platform

### Advanced E-Commerce with Vendor Logins & Payment Integration

**Project Owner:** Arnold Marten
**Project Duration:** Weeks 9-15 (Spring 2026)
**Total Estimated Hours:** 48 - 52 hours
**Spring Break (Week 9):** Optional work window
**Active Development:** Weeks 10-15 (6 weeks)

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## TIMELINE OVERVIEW

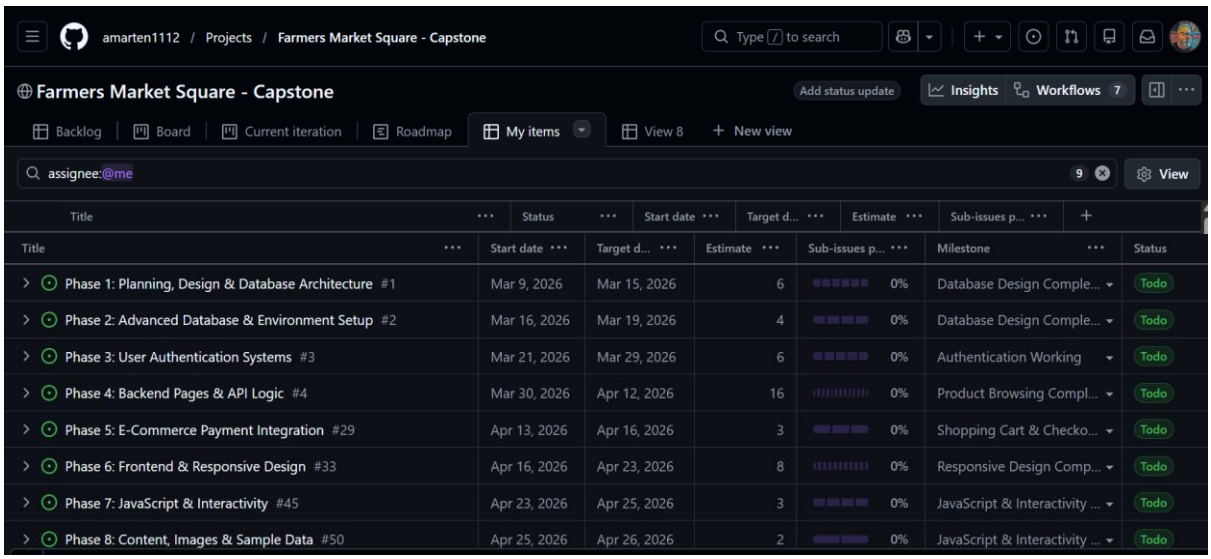


*Figure 1 - Screenshot from GitHub Project*

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# PHASE BREAKDOWN WITH DETAILED TASKS

## PHASE 1: PLANNING, DESIGN & DATABASE ARCHITECTURE

**Total Hours: 6 | Status: Upcoming**

### Phase 1 Overview

Establish project scope, create comprehensive design documentation, and develop advanced database architecture for the multi-user e-commerce system supporting customers, vendors, and administrators.

### Phase 1 Tasks

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 1: Planning, Design & Database Architecture #1 | Todo | Mar 9, 2026 | Mar 15, 2026 | 6 | 0% |
| 1.1 Project charter with scope and requirements #57 | Todo | Mar 9, 2026 | Mar 10, 2026 | 1 | |
| 1.2 Sitemap with 3 user roles (customer, vendor, a... #58 | Todo | Mar 9, 2026 | Mar 10, 2026 | 1 | |
| 1.3 20+ page wireframes #59 | Todo | Mar 10, 2026 | Mar 11, 2026 | 1 | |
| 1.4 Visual mockups of key pages #60 | Todo | Mar 11, 2026 | Mar 12, 2026 | 1 | |
| 1.5 Complete ER (Entity Relationship) diagram #61 | Todo | Mar 11, 2026 | Mar 12, 2026 | 1 | |
| 1.6 Database design document with normalization ... #62 | Todo | Mar 13, 2026 | Mar 14, 2026 | 1 | |

*Figure 2 - Screenshot from GitHub Project*

**Phase 1 Deliverables:**

- ☐Project charter document
- ☐User flow diagrams (Customer, Vendor, Admin)
- ☐Wireframes (20+ pages)
- ☐Visual mockups (5+ key pages)
- ☐ER diagram with 12-table schema
- ☐Database design document

**Phase 1 Success Criteria:**

- ✓ All documentation reviewed and approved
- ✓ Database design normalized to 3NF
- ✓ Ready to begin Phase 2 implementation

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## PHASE 2: ADVANCED DATABASE & ENVIRONMENT SETUP

**Total Hours: 4 | Status: Upcoming**

## Phase 2 Overview

Create normalized database schema with 12 interconnected tables, establish development environment with XAMPP/Git, and populate with comprehensive test data.

## Phase 2 Tasks

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 2: Advanced Database & Environment Setup #2 | Todo | Mar 16, 2026 | Mar 19, 2026 | 4 | 0% |
| 2.1 Design normalized relational schema #7 | Todo | Mar 16, 2026 | Mar 19, 2026 | 1 | |
| 2.2 Write SQL schema & create tables #8 | Todo | Mar 16, 2026 | Mar 19, 2026 | 1.5 | |
| 2.3 Set up development environment #9 | Todo | Mar 16, 2026 | Mar 19, 2026 | 0.5 | |
| 2.4 Populate test data & fixtures #10 | Todo | Mar 16, 2026 | Mar 19, 2026 | 1 | |

*Figure 3 Screenshot GitHub Project*

### Phase 2 Deliverables:

- ☐database.sql with 12 tables
- ☐Complete test database (local)
- ☐ER diagram (visual export)
- ☐Database documentation
- ☐Development environment verified

### Phase 2 Success Criteria:

- ✓ Database creates without errors
- ✓ All relationships (foreign keys) functional
- ✓ Test data populates successfully
- ✓ Ready for Phase 3

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## PHASE 3: USER AUTHENTICATION SYSTEMS

**Total Hours: 6 | Status: Upcoming**

### Phase 3 Overview

Implement secure, role-based authentication systems for three distinct user types (Admin, Vendor, Customer) with session management, password hashing, and CSRF protection.

### Phase 3 Tasks

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 3: User Authentication Systems #3 | Todo | Mar 21, 2026 | Mar 29, 2026 | 6 | 0% |
| 3.1 Admin authentication system #11 | Todo | Mar 21, 2026 | Mar 22, 2026 | 1 | |
| 3.2 Vendor authentication system #12 | Todo | Mar 22, 2026 | Mar 23, 2026 | 1 | |
| 3.3 Customer authentication system #13 | Todo | Mar 24, 2026 | Mar 24, 2026 | 1.5 | |
| 3.4 Authentication helper functions #14 | Todo | Mar 26, 2026 | Mar 25, 2026 | 1 | |
| 3.5 Session security & logout #15 | Todo | Mar 27, 2026 | Mar 28, 2026 | 1 | |

*Figure 4 Screenshot GitHub Project*

**Phase 3 Deliverables:**

- ☐Admin login page & system
- ☐Vendor login & registration approval workflow
- ☐Customer login & registration page
- ☐Password hashing implementation (PHP password_hash)
- ☐Session validation & CSRF token generation
- ☐Helper functions library (check_login, check_vendor, etc.)

**Phase 3 Success Criteria:**

- ✓ All 3 user types can login successfully
- ✓ Passwords stored hashed (never plain text)
- ✓ Session timeouts working correctly
- ✓ Role-based access control verified
- ✓ Ready for Phase 4

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# PHASE 4: BACKEND PAGES & API LOGIC

## Total Hours: 14 | Status: Upcoming

## Phase 4A: Product Browsing & Discovery

### Sub-Phase Hours: 6

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 4: Backend Pages & API Logic #4 | Todo | Mar 30, 2026 | Apr 12, 2026 | 16 | 0% |
| 4.1 Product browsing pages (with JOINs) #19 | Todo | Mar 30, 2026 | Mar 31, 2026 | 2 | |
| 4.2 Category filtering & search #20 | Todo | Apr 2, 2026 | Apr 3, 2026 | 1.5 | |
| 4.3 Individual product detail page #21 | Todo | Apr 3, 2026 | Apr 4, 2026 | 1 | |
| 4.4 Vendor storefront pages #22 | Todo | Apr 4, 2026 | Apr 5, 2026 | 1 | |
| 4.5 Contact & event pages #23 | Todo | Apr 5, 2026 | Apr 6, 2026 | 0.5 | |

*Figure 5 Screenshot GitHub Project*

## Phase 4B: Shopping Cart & Checkout

### Sub-Phase Hours: 8

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| 4.6 Add to cart (cart table inserts) #16 | Todo | Apr 6, 2026 | Apr 8, 2026 | 1 | |
| 4.7 View cart page (with JOINs for totals) #17 | Todo | Apr 7, 2026 | Apr 9, 2026 | 1 | |
| 4.8 Update cart quantities, remove items #18 | Todo | Apr 8, 2026 | Apr 10, 2026 | 1 | |
| 4.9 Checkout flow (multi-step) #24 | Todo | Apr 9, 2026 | Apr 10, 2026 | 1.5 | |
| 4.10 Order creation & confirmation #25 | Todo | Apr 10, 2026 | Apr 11, 2026 | 1 | |
| 4.11 Order history & tracking #26 | Todo | Apr 10, 2026 | Apr 11, 2026 | 0.5 | |
| 4.12 Vendor dashboard (sales metrics with JOINs) #27 | Todo | Apr 11, 2026 | Apr 12, 2026 | 1 | |
| 4.13 Vendor product management #28 | Todo | Apr 11, 2026 | Apr 12, 2026 | 1 | |

*Figure 6 Screenshot GitHub Project*

### Phase 4 Deliverables:

- ☐Product listing with pagination
- ☐Category filtering system
- ☐Search functionality
- ☐Product detail pages (with images, vendor info)
- ☐Vendor storefront pages
- ☐Shopping cart (database-backed)
- ☐5-step checkout process
- ☐Order creation & management
- ☐Order history/tracking for customers
- ☐Vendor dashboard with sales metrics
- ☐Vendor product management interface

### Phase 4 Success Criteria:

- ✓ All pages display correctly with data from JOINs
- ✓ Cart functionality working (add, view, update, remove)
- ✓ Checkout flow complete (all 5 steps)
- ✓ Orders created and stored correctly
- ✓ Vendor dashboard shows accurate data
- ✓ Ready for Phase 5

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## PHASE 5: FRONTEND & RESPONSIVE DESIGN

**Total Hours: 15 | Status: Upcoming**

### Phase 5 Overview

Build professional, responsive user interface using Bootstrap 5 framework with custom styling, component customization, and comprehensive mobile testing across all devices.

### Phase 5 Tasks

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 6: Frontend & Responsive Design #33 | Todo | Apr 16, 2026 | Apr 23, 2026 | 8 | 0% |
| 6.1 Bootstrap framework setup & customization #34 | Todo | Apr 16, 2026 | Apr 17, 2026 | 1.5 | |
| 6.2 Homepage responsive design #35 | Todo | Apr 16, 2026 | Apr 17, 2026 | 1 | |
| 6.3 Product catalog & filtering UI #36 | Todo | Apr 16, 2026 | Apr 17, 2026 | 2 | |
| 6.4 Product detail page styling #37 | Todo | Apr 17, 2026 | Apr 18, 2026 | 1.5 | |
| 6.5 Shopping cart page styling #38 | Todo | Apr 18, 2026 | Apr 19, 2026 | 1.5 | |
| 6.6 Multi-step checkout UI #39 | Todo | Apr 18, 2026 | Apr 19, 2026 | 2 | |
| 6.7 Vendor dashboard styling #40 | Todo | Apr 18, 2026 | Apr 19, 2026 | 1.5 | |
| 6.8 Forms styling (all forms) #41 | Todo | Apr 20, 2026 | Apr 21, 2026 | 1.5 | |
| 6.9 Navigation & layouts #42 | Todo | Apr 20, 2026 | Apr 22, 2026 | 1 | |
| 6.10 Mobile responsiveness testing #43 | Todo | Apr 20, 2026 | Apr 23, 2026 | 1 | |
| 6.11 Cross-browser testing #44 | Todo | Apr 23, 2026 | Apr 23, 2026 | 1 | |

*Figure 7 Screenshot GitHub Project*

**Phase 5 Deliverables:**

- ☐Bootstrap 5 fully integrated
- ☐Custom CSS styling complete
- ☐All pages responsive (mobile-first)
- ☐Homepage with hero & featured products
- ☐Product catalog grid
- ☐Product detail page styling
- ☐Shopping cart responsive layout
- ☐Multi-step checkout styling
- ☐Vendor dashboard responsive layout
- ☐All forms styled and validated
- ☐Responsive navbar & footer
- ☐Mobile menu functionality

**Phase 5 Success Criteria:**

- ✓ Mobile (375px - 480px) renders correctly
- ✓ Tablet (768px - 1024px) renders correctly

- ✓ Desktop (1200px+) renders correctly
- ✓ Cross-browser compatibility verified (Chrome, Firefox, Safari, Edge)
- ✓ Touch-friendly buttons on mobile
- ✓ Ready for Phase 7

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## PHASE 6: E-COMMERCE PAYMENT INTEGRATION

**Total Hours: 3 | Status: Upcoming**

### Phase 6 Overview

Integrate Stripe payment processing API with comprehensive error handling, transaction logging, and secure payment workflows in test mode.

### Phase 5 Tasks

| Title | Status | Start date | Target d... | Estimate | Sub-issues p... |
|---|---|---|---|---|---|
| Phase 5: E-Commerce Payment Integration #29 | Todo | Apr 13, 2026 | Apr 16, 2026 | 3 | 0% |
| 5.1 Stripe API setup & testing #30 | Todo | Apr 13, 2026 | Apr 14, 2026 | 1 | |
| 5.2 Payment processing page #31 | Todo | Apr 14, 2026 | Apr 15, 2026 | 1 | |
| 5.3 Payment error handling #32 | Todo | Apr 15, 2026 | Apr 16, 2026 | 1 | |

*Figure 8 Screenshot GitHub Project*

**Phase 6 Deliverables:**

- ☐Stripe account configured with test API keys
- ☐Payment form with Stripe.js integration
- ☐Charge creation & processing
- ☐Payment success page
- ☐Payment failure page with retry logic
- ☐Transaction logging system

**Phase 6 Success Criteria:**

- ✓ Stripe test keys configured and working
- ✓ Test payment cards process successfully
- ✓ Failed card handling displays error messages
- ✓ Transaction logging functional
- ✓ Ready for Phase 6

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## PHASE 7: JAVASCRIPT & INTERACTIVITY

**Total Hours: 3 | Status: Upcoming**

### Phase 7 Overview

Add client-side validation, interactive features, AJAX cart updates, and performance optimizations for enhanced user experience.

### Phase 7 Tasks

| | | | | | |
|---|---|---|---|---|---|
| Phase 7: JavaScript & Interactivity #45 | Todo | Apr 23, 2026 | Apr 25, 2026 | 3 | 0% |
| 7.1 Form validation (client-side) #46 | Todo | Apr 23, 2026 | Apr 25, 2026 | 1 | |
| 7.2 Shopping cart interactivity #47 | Todo | Apr 23, 2026 | Apr 25, 2026 | 1 | |
| 7.3 Mobile menu & interactive elements #48 | Todo | Apr 23, 2026 | Apr 25, 2026 | 1 | |
| 7.4 Image handling & optimization #49 | Todo | Apr 23, 2026 | Apr 25, 2026 | 1 | |

*Figure 9 Screenshot GitHub Project*

**Phase 7 Deliverables:**

- ☐Form validation (real-time feedback)
- ☐Cart quantity updates via AJAX
- ☐Remove items via AJAX
- ☐Smooth animations
- ☐Mobile hamburger menu
- ☐Image lazy loading
- ☐Progress indicators

**Phase 7 Success Criteria:**

- ✓ All forms validated before submission
- ✓ Cart updates without page reload
- ✓ Smooth user experience
- ✓ Ready for Phase 8

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## PHASE 8: CONTENT, IMAGES & SAMPLE DATA

**Total Hours: 2 | Status: Upcoming**

### Phase 8 Overview

Create realistic sample content, populate database with vendor profiles, products, and orders, and optimize all media assets.

### Phase 8 Tasks

| | | | | | |
|---|---|---|---|---|---|
| Phase 8: Content, Images & Sample Data #50 | Todo | Apr 25, 2026 | Apr 26, 2026 | 2 | 0% |
| 8.1 Create sample vendor & product content #51 | Todo | Apr 26, 2026 | Apr 26, 2026 | 1 | |
| 8.2 Gather & optimize images #52 | Todo | Apr 26, 2026 | Apr 26, 2026 | 1 | |

*Figure 10 Screenshote GitHub Project*

**Phase 8 Deliverables:**

- ☐3 - 5 vendor profiles
- ☐10-20 products with descriptions
- ☐Product images (optimized)
- ☐Realistic database content
- ☐Sample events
- ☐Sample orders for testing

**Phase 8 Success Criteria:**

- ✓ Database populated with realistic data
- ✓ Images optimized for web
- ✓ Ready for Phase 9

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## PHASE 9: TESTING, OPTIMIZATION & DEPLOYMENT

**Total Hours: 3 | Status: Upcoming**

### Phase 9 Overview

Comprehensive testing of all features, security review, performance optimization, bug fixes, and production deployment.

### Phase 9 Tasks

| | | | | | |
|---|---|---|---|---|---|
| Phase 9: Testing, Optimization & Deployment #53 | Todo | Apr 25, 2026 | Apr 26, 2026 | 3 | 0% |
| 9.1 Comprehensive testing (all features) #54 | Todo | Apr 25, 2026 | Apr 26, 2026 | 1 | |
| 9.2 Bug fixes & security review #55 | Todo | Apr 25, 2026 | Apr 26, 2026 | 1 | |
| 9.3 Hosting setup & deployment #56 | Todo | Apr 25, 2026 | Apr 26, 2026 | 1 | |

*Figure 11 Screenshot GitHub Project*

**Phase 9 Deliverables:**

- ☐Test report (all features working)
- ☐Security audit completed
- ☐Bug fixes documented
- ☐Live website on production server
- ☐SSL certificate installed
- ☐Deployment documentation
- ☐Admin & vendor user manuals

**Phase 9 Success Criteria:**

- ✓ All login flows working
- ✓ Product browsing & search functional
- ✓ Shopping cart & checkout complete
- ✓ Payment processing working (Stripe test mode)
- ✓ No SQL injection vulnerabilities
- ✓ No XSS vulnerabilities
- ✓ Website live and accessible
- ✓ **Project Complete**

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# 📋 TESTING CHECKLIST (Phase 9)

## Functional Testing

- ☐Admin login/logout working
- ☐Vendor login/logout working
- ☐Customer login/logout working
- ☐Product browsing & search functional
- ☐Add to cart functionality working
- ☐View cart displaying correct items
- ☐Update quantities in cart
- ☐Remove items from cart
- ☐Checkout process (all 5 steps)
- ☐Order confirmation received
- ☐Order history displaying
- ☐Vendor dashboard showing metrics
- ☐Vendor product management working

## Security Testing

- ☐No SQL injection vulnerabilities
- ☐No XSS vulnerabilities
- ☐Passwords stored hashed (verified)
- ☐Session security verified
- ☐Role-based access control working

## Responsive Testing

- ☐Mobile (375px - 480px) - All pages render
- ☐Tablet (768px - 1024px) - All pages render
- ☐Desktop (1200px+) - All pages render
- ☐Chrome browser compatibility
- ☐Firefox browser compatibility
- ☐Safari browser compatibility
- ☐Edge browser compatibility

## Performance Testing

- ☐Page load times < 3 seconds

- ☐Images optimized (< 1MB each)
- ☐Database queries optimized
- ☐No broken links

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## 🚑 RISK MANAGEMENT & CONTINGENCY PLANNING

### Risk 1: Falling Behind Schedule

**Probability:** Medium | **Impact:** High

- **Mitigation:**
  - Use Bootstrap templates to speed up styling
  - Reduce vendor dashboard complexity if needed
  - Defer advanced search filters to Phase 2
- **Contingency:** Deploy with essential features only by Week 15 end

### Risk 2: Database Complexity Issues

**Probability:** Low | **Impact:** High

- **Mitigation:**
  - Spend extra time in Phase 1 on database design
  - Test all JOINs in Phase 2
  - Have database expert review schema
- **Contingency:** Simplify schema to 8 tables instead of 12

### Risk 3: Payment Integration Delays

**Probability:** Low | **Impact:** Medium

- **Mitigation:**
  - Start Stripe setup early in Phase 5
  - Use Stripe's test mode thoroughly
  - Have backup payment solution (PayPal)
- **Contingency:** Deploy without payment initially, add later

### Risk 4: Mobile Responsiveness Issues

**Probability:** Medium | **Impact:** Medium

- **Mitigation:**
  - Test on actual devices throughout project
  - Use Bootstrap's responsive grid system
  - Start mobile-first approach
- **Contingency:** Simplify Design

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## 📞 SUPPORT RESOURCES

**Internal Resources:**

- GitHub Repository: amarten1112/Capstone
- Documentation: See Phase 1 deliverables
- Code Examples: Available in repository

**External Resources:**

- PHP Documentation: php.net
- Bootstrap Documentation: getbootstrap.com
- Stripe Documentation: stripe.com/docs
- MDN Web Docs: developer.mozilla.org

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**Report Generated:** March 1, 2026
**Project Status:** In Planning Phase
**Next Review:** End of Week 10

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*This PM report is a living document and will be updated weekly with actual progress, revised timelines, and any scope changes.*